

SUMMARY

I recently graduated in Electrical and Electronic Engineering. Being passionate about research and problem-solving, I am seeking a PhD program and a Graduate Assistantship to deepen my involvement in research and contribute to academic excellence. My goal is to make a significant impact in application field of medical wearable devices and AI, leveraging my research experience, teamwork, and problem-solving skills for innovative projects.

ACADEMIC CREDENTIALS

•Rajshahi University of Engineering & Technology

CGPA: 3.72/4.00

B.Sc. in Electrical and Electronic Engineering

March, 2022 – September, 2026

RESEARCH INTERESTS

Biomedical Signal Processing (ECG, PPG), Non-Invasive Biosensing, Deep Learning & Model Optimization, Applied Machine Learning & Cross-Domain Modeling (Mining Engineering Applications).

PROFESSIONAL EXPERIENCE

•SPML Research Lab, RUET

Sep 2026 – Present

Research Assistant

Rajshahi, Bangladesh

– Constructing a multimodal biomedical dataset by ECG, PPG, and GSR signals using Shimmer sensor devices.

Undergraduate Thesis Researcher

Sep 2024 – Aug 2026

- Developed deep learning frameworks for Bundle Branch Block
- Designed non-invasive blood glucose level prediction PPG signals.

•ResearchBuddy AI Lab, CUET

Research Trainee

Jan, 2024 – June, 2024

– Explored LLM and self-supervised learning (SSL) models.

•Udvash Academic & Admission Care

Academic Evaluator & Tutor

Mar, 2022 – Dec, 2023

– Evaluated exam scripts and provided academic guidance to students on difficult topics.

PUBLICATIONS

Journal Articles

1. **T Ahmmod**, MA Motin, and S Kabir, “A Generalizable Framework for Bundle Branch Block Detection Using Knowledge Distillation-Guided Lead-Aware Convolutional Neural Networks and Hybrid Feature Fusion,” *IEEE Transactions on Human-Machine Systems*. (Under Review) [Field: Bio-ML]

Conference Proceedings

1. **T Ahmmod**, MT Amin, Z Ferdus, TNH Naem, E Ferdous, and MM Tajwear, “Cross-Group Meta-Ensemble Learning for Heart Disease Classification Using ML, DL, SSL, and Biomedical PLMs Models,” in *Proc. IEEE SPICSCON*, 2026. (Accepted) [Field: Bio-ML]
2. **T Ahmmod***, R Sarkar*, and MA Motin, “GlucoNova: A Hybrid Model for Non-Invasive Blood Glucose Monitoring Using Near-Infrared Photoplethysmography,” in *Proc. 28th International Conference on Computer and Information Technology (ICCIT)*, 2025. DOI: 10.1109/ICCIT68739.2025.11490406. [Field: Bio-Device]
3. MT Amin*, **T Ahmmod***, and TNH Naem, “Integrating Machine Learning Ensembles and Large Language Models for Heart Disease Prediction Using Voting Fusion,” in *Proc. ICECTE*, 2026. [Field: Bio-ML]
DOI: <https://doi.org/10.1109/ICECTE69292.2026.11429383>.
4. A Chaudhury and **T Ahmmod**, “Point-Supervised Cross-Conditioned Cascade Framework for Explainable Joint Tumor and Modality Classification in Brain MRI,” in *Proc. 29th International Conference on Computer and Information Technology (ICCIT)*, 2026. (Under Review) [Field: Bio-ML]
5. A Deb, **T Ahmmod**, and S Siddique, “Empirical-Driven Machine Learning and Bayesian Optimization for Uniaxial Compressive Strength Prediction from Bengal Basin,” in *Proc. 29th International Conference on Computer and Information Technology (ICCIT)*, 2026. (Under Review) [Field: Cross-ML]

* indicates equal contribution

PROJECTS & WORKS

•Multi Signal Database

RUET TechnoFair, 2025

–Description: Developed a deep-learning powered embedded software designed to analyze multi-channel ECG signals, automatically detecting arrhythmias with high precision using the PyTorch framework.

–Tools: Python, PyTorch, Signal Processing, Deep Learning.

•Iontophoresis Device for Hyperhidrosis Treatment

RUET Project 1, 2024

–Description: Designed an automated, safe medical device for non-invasive treatment of hyperhidrosis (excessive sweating).

–Tools: Embedded Systems, Circuit Design, Medical Device, Hyperhidrosis Therapy.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, MATLAB, HTML

Python Libraries & Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, Keras, PyTorch

Hardware & Embedded Systems: Arduino, Shimmer ECG/PPG/GSR Unit, Multi-Sensors, Proteus, Verilog

Engineering Tools & Software: EasyEDA, Arduino IDE, Kaggle, AutoCAD

Documentation & Design: LaTeX, MS Office (Word, Excel, PowerPoint), Canva

Research Highlights: First Author (2 Journals, 3 Conferences), Attended 3 Conferences

ACHIEVEMENTS & MEMBERSHIPS

• RUET Merit Scholarship

2022 – 2026

Awarded for outstanding academic excellence and maintaining a high CGPA.

• IEEE Student Membership

2022 – Present

Active student member of the Institute of Electrical and Electronics Engineers (IEEE).

• ASSR Membership

Club Member

Active member of Astronomy and Science Society of RUET (ASSR).

REFERENCES

Dr. Mohammad Abdul Motin

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